

Sajeevan Veeriah

Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer

Geelong, Victoria, Australia | +61 498 586 654 | sajeevanveeriah@gmail.com

sajeevanveeriah.github.io | linkedin.com/in/sajeevan-veeriah | github.com/Sajeevanveeriah



01 Professional Profile

I engineer complete systems across the physical, control, digital and operational boundary, then verify them as one working system. My practice spans robotics, industrial automation, embedded systems, industrial IT/OT, AI/ML and engineering software from requirements through integration, commissioning and handover.

02 Incoming Appointment

INCOMING APPOINTMENT - CONTRACT PENDING

Industrial IT Engineer - IT Systems / MES / ERP

Farm Frites Australia | Dooen, Victoria

Planned appointment. Employment has not commenced; the final public title and commencement date remain subject to the issued contract.

- Role scope includes MES implementation and production workflows, ERP integration, SCADA/PLC/historian interfaces, OT infrastructure and production-data integrity.
- The position also covers commissioning, validation, traceability, vendor coordination, cybersecurity, documentation and cross-functional support across Operations, Engineering, Quality, Supply Chain and IT.

03 Core Expertise

Industrial IT, OT and Automation PLC and HMI/SCADA integration, MES exposure, production-data interfaces, field devices, drives, industrial networks, FAT, SAT and commissioning.	Robotics and Mechatronics ROS 2 Humble, Nav2, MoveIt 2, Gazebo Fortress, SLAM, EKF sensor fusion, localisation, planning, motion control and physical prototyping.
Embedded Systems and Connectivity ESP32, STM32, FreeRTOS, C/C++, PCB bring-up, CAN, UART, I2C, SPI, BLE, LoRaWAN, MQTT and Linux services.	AI/ML, Data and Engineering Software Python, scikit-learn, OpenCV, YOLO, time-series analysis, MATLAB, Simulink, TypeScript, React, Rust, SQL and CI/CD.
Verification and Delivery Requirements, instrumentation, calibrated testing, regression, fault isolation, data integrity, traceable QA, validation and technical handover.	Manufacturing and Quality GMP, food and beverage operations, traceability, batch discipline, process troubleshooting, ITP/MDR records, WHS and regulated evidence.

04 Professional Experience

Automation and Controls Engineer

Jan 2026 - Jun 2026

Process automation systems integrator

Regulated pharmaceutical, biotechnology and food production

- Delivered PLC logic, HMI/SCADA, field-device, drive, MES and production-data integration under GMP controls.
- Migrated validated application content from iFIX to PVI+, completed functional checks, and supported FAT, SAT, commissioning, handover and MiR mobile-robot fault tracing.

Product Development Test Engineer (Contract)

Oct 2025 - Jan 2026

Global automotive OEM via engineering consultancy

Vehicle software integration and ADAS product development

- Validated vehicle-software integration and ADAS behaviour through feature-vehicle, breadboard and regression testing.
- Instrumented test vehicles and used Vector CANoe and CANalyzer to capture CAN and CAN FD evidence for fault isolation and software-readiness verification.



04 Professional Experience Continued

Technical Officer and Quality Assurance

Jul 2024 - Aug 2025

ABMARC

Automotive testing, energy, emissions and compliance engineering

- Delivered ADR and EURO emissions work, EV/PHEV range testing and vehicle-systems evaluation using calibrated instrumentation, data acquisition and CAN tools.
- Maintained traceable QA, safety, regulatory and test records for certification review and audit.

Consultant Engineer, IoT and Projects Administrator

Feb 2024 - Aug 2024

DuxTel

Networking, low-power telemetry and remote asset visibility

- Built field-to-dashboard solutions with ESP32, sensors, LoRaWAN AU915, MQTT, ChirpStack, InfluxDB and Grafana.
- Supported CAN and GPS capture, custom PCB work, MikroTik connectivity and Linux services from field trial through handover.

Manufacturing, Commissioning and Quality Foundation

IDL Australia | Carbon Revolution | Thornton Engineering Australia

2018 - 2024 | Food and beverage, carbon-fibre and structural-steel production

- Built practical experience in equipment operation, changeovers, safe production, first-response fault recovery, process discipline, batch quality and traceability.
- Supported drawing review, ITP/MDR documentation, material traceability, QA inspection, equipment trials and commissioning checks across workshop and production environments.

05 Selected Engineering Projects

Open Industrial Automation

2026 | Public industrial reference and simulation platform

Built a deterministic mixing, dosing and CIP reference process with an operator HMI, Engineering Studio, typed project model, traceability, cybersecurity boundaries and automated verification gates.

Autonomous Navigation Rover on ROS 2

2024 - 2025 | Hardware build with simulation-validated autonomy

Built the differential-drive platform and integrated LiDAR, IMU, SLAM, EKF state estimation, Nav2 planning, control and recovery across independently testable ROS 2 nodes.

ESP32 Clinical Ataxia Assessment Device

2025 | Honours capstone and assessed embedded prototype

Designed the device, custom PCB and enclosure, implemented deterministic 100 Hz acquisition, Bluetooth live display and CSV/PDF reporting, then checked accuracy, reversal, drift and temperature behaviour against reference instruments.

06 Education

Bachelor of Mechatronics Engineering (Honours), Distinction

Deakin University, Geelong | 2025 | Honours capstone: ESP32 clinical ataxia assessment device.

Higher National Diploma in Mechatronics, Robotics and Automation Engineering, Distinction

Cardiff Metropolitan University, United Kingdom | 2016.

07 Professional Membership

Member, Engineers Australia

08 Professional Development

Lean Six Sigma Foundation | JIRA and Agile | KAIZEN | Industrial Automation and IIoT | AI/ML | CAD

09 Languages and Additional Information

Languages: English, Tamil and Sinhala.

Additional: Current Victorian driver licence | Community leadership through Newcomb and District Cricket Club | Engineering mentoring and peer support.